THE DIAGONAL OF A LOCALLY OF FINITE TYPE MORPHISM OF ALGEBRAIC STACKS IS LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The base change permanence of being locally of finite type. Invoked inside the counted proof (source lines 5717-5719). Makes the second projection from the fibre product locally of finite type over the source.
- The permanence statement for finite presentation for a morphism factored through a finite presentation morphism over the source. Invoked inside the counted proof (source lines 5720-5721). Turns local finite type of the diagonal over the source into local finite presentation.

Source reference. Original lemma lemma-base-change-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3760–3760. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-finite-presentation-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5721–5721. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-presentation-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks with diagonal $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$. If f is locally of finite type then Δ is locally of finite presentation. If f is quasi-separated and locally of finite type, then Δ is of finite presentation.*

Proof. Note that Δ is a morphism over $\mathcal{X}$ (via the second projection). If f is locally of finite type, then $\mathcal{X}$ is of finite presentation over $\mathcal{X}$ and $\mathrm{pr}_2 : \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{X}$ is locally of finite type by Lemma lemma-base-change-finite-type. Thus the first statement holds by Lemma lemma-finite-presentation-permanence. The second statement follows from the first and the definitions (because f being quasi-separated means by definition that Δ_f is quasi-compact and quasi-separated). $\square$